

```
In [1]: def print_triangle(n):
        pass
        for i in range(1,6):
            for j in range(1,i+1):
                print('*',end='')
            print()
        print(          )

        def print_square(n):
            pass
            for i in range(1,6):
                for j in range(1,6):
                    print('*',end='')
                print()
            print(          )
        def print_reverse(n):
            pass
            for i in range(1,6):
                for j in range(1,7-i):
                    print('*',end='')
                print()

        print_triangle(5)
        print()
        print_square(4)
        print()
        print_reverse(5)
```

```
*
**
***
****
*****

*****
*****
*****
*****
*****

*****
****
***
**
*
```

```

In [2]: g = [
    ["Rahul", 78, 85, 72],
    ["Priya", 92, 88, 95],
    ["Arjun", 45, 38, 62],
    ["Sneha", 88, 91, 80],
    ["Vikram", 55, 48, 70],
    ]
subjects = ["Maths", "Science", "English"]
t=[]
tp=0
top=[]
fl=[]
av1=0
av2=0
av3=0
s=0
d=0
f=0
for i,j,k,l in g:
    t.append((i,j+k+l))
for i,j in t:
    if j>tp:
        tp=j
print(t)
for i,j in t:
    if tp==j:
        top.append((i,j))
print('Class topper :',top)
for i,j,k,l in g:
    if j<50 or k<50 or l<50:
        fl.append(i)
print('Failed students :',fl)
for i,j,k,l in g:
    s=s+j
    d=d+k
    f=f+l

av1=s/5
av2=d/5
av3=f/5
print('subjects average:','maths',av1,'Science',av2,'English',av3)

[('Rahul', 235), ('Priya', 275), ('Arjun', 145), ('Sneha', 259), ('Vikram', 173)]
Class topper : [('Priya', 275)]
Failed students : ['Arjun', 'Vikram']
subjects average:maths 71.6 Science 70.0 English 75.8

```

In []: